Exact characterisation and certification of the sharp two-sided Cheeger–Buser bound for finite weighted graphs

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Abstract

Cheeger's inequality and Buser's inequality are the classical two-sided comparison between the second eigenvalue of the normalised Laplacian of a graph and its Cheeger constant. Both are usually quoted for regular, loop-free graphs, and implementations that apply them more broadly are easy to get subtly wrong: we document three distinct defects that arose in our own implementation. We give a self-contained statement of the two-sided bound for arbitrary finite *weighted* graphs with self-loops, with hypotheses spelled out precisely enough that the statement is actually true, and we work through its proof by the co-area method rather than quoting the conclusion. We then determine the extremal coefficient exactly: for the two-vertex family with equal self-loop weight, $\lambda_2/h = 2$ identically, so the constant 2 in Buser's bound is attained and cannot be lowered. Finally we build a verification pipeline that checks both inequalities on *every* connected graph on up to six vertices and on weighted graphs with self-loops up to four vertices using exact rational arithmetic and exact characteristic polynomials; it finds no violation, and it correctly reproduces the known equality case. We report the pipeline's corrections, including a defect that had produced a false counterexample in an earlier draft. The inequalities themselves are classical and we do not claim them as new; the contribution is the exact characterisation, the certified computation, and the reproducible verification.

Keywords: spectral graph theory; Cheeger inequality; Buser inequality; normalised Laplacian; isoperimetric constant; exact computation; computer-assisted verification.

1 Introduction

Fix a finite graph G and let λ_2 be the second-smallest eigenvalue of its normalised Laplacian. The Cheeger constant h measures how badly G can be cut. The classical message is that λ_2 and h control one another:

$$1 - \sqrt{h(2-h)} \;\le\; \lambda_2 \;\le\; 2h + 2h^2, \tag{1}$$

the left-hand side being Cheeger's bound and the right-hand side Buser's [1–5]. The bound (1) is the engine behind spectral clustering, expander construction, mixing-time estimates and, in higher-dimensional form, the higher-order Cheeger inequalities of [6].

Equation (1) is usually stated for d-regular, loop-free graphs. This is not pedantry: both the constant and the meaning of h depend on the degree normalisation, and a statement that is silently false outside the regular case is a genuine trap. In the course of this work we built such an implementation, and it produced three separate errors, one of which manufactured a convincing but entirely spurious "counterexample" to Buser's inequality for graphs with self-loops. We report all three, because they are instructive and because the point of the paper is

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that the bound survives a much wider class than it is usually quoted for — but only if every hypothesis is stated.

Our contributions are as follows.

1. **A precisely stated two-sided theorem** (Theorem 4.2) for arbitrary finite weighted graphs with self-loops, proved from scratch in Section 4.
2. **An exact extremal characterisation** (Theorem 3.1) of the optimal coefficient in the reverse direction: for the family $W = \begin{pmatrix} a & w \\ w & a \end{pmatrix}$ with $a, w > 0$ we have $\lambda_2/h = 2$ identically, so the constant 2 is attained, not merely approached, and cannot be improved.
3. **A certified verification pipeline** that checks both inequalities on every connected graph with $n \leq 6$ vertices and on weighted graphs with self-loops with $n \leq 4$, using exact rational arithmetic throughout (Section 5).
4. **A candid account of three implementation defects** discovered by that pipeline, including one that had produced a false counterexample in an earlier version of this work (Section 6).

We are explicit that (1) is classical. What is new here is the exact characterisation of the extremal constant, the certified computation, and the reproducible verification; see Section 7 and Section 8.

2 Definitions and conventions

Let $G = (V, W)$ be a finite connected *weighted graph* on $n = |V| \geq 2$ vertices: W is a symmetric matrix of non-negative real numbers, $W_{vu} = W_{uv}$, and W_{vv} is the weight of a self-loop at v. Throughout we assume:

(A1) $W_{vv} \geq 0$, $W_{vu} = W_{uv}$, and $W_{vu} \geq 0$;

(A2) the degree $d_v = \sum_{u \in V} W_{vu}$ is strictly positive for every v;

(A3) the graph is connected, in the sense that the support of the off-diagonal part of W spans a connected graph on V.

Definition 2.1 (Volume, boundary, Cheeger constant)**.** For $S \subseteq V$ write

$$\mathrm{vol}(S) = \sum_{v \in S} d_v, \qquad \mathrm{bnd}(S) = \sum_{v \in S,\, u \in V \setminus S} W_{vu},$$

and define

$$h(G) = \min \Big\{ \frac{\mathrm{bnd}(S)}{\mathrm{vol}(S)} \;:\; \emptyset \neq S \subsetneq V,\ \mathrm{vol}(S) \leq \tfrac{1}{2}\,\mathrm{vol}(V) \Big\}.$$

Note that $\mathrm{bnd}(S)$ counts only edges that *leave* S. A self-loop has both endpoints in S (or neither), so self-loops never appear in $\mathrm{bnd}(S)$. One checks directly that

$$\mathrm{bnd}(S) = \mathrm{vol}(S) - 2E(S) - \Lambda(S), \tag{2}$$

where $E(S) = \sum_{v<w,\, v,w \in S} W_{vw}$ is the total weight of edges internal to S and $\Lambda(S) = \sum_{v \in S} W_{vv}$ is the total weight of self-loops in S; this identity is used as a cross-check in the test suite. Since $\mathrm{vol}(S) \geq \mathrm{bnd}(S)$, we always have $h(G) \in (0, 1]$.

Remark 2.2. For a d-regular loop-free graph, $\mathrm{vol}(S) = d|S|$, so $h(G) = \min_S |E(S, S^c)|/(d|S|)$ with $|S| \le n/2$, which is exactly the normalisation in which the classical statements of [1] are written; note $2h+2h^2 = 2h(1+h)$ and $1-\sqrt{h(2-h)} = 1-\sqrt{2h(1-h/2)}$. Because $\mathrm{vol}(S)$ and $\mathrm{vol}(V \setminus S)$ agree only in the regular case, the complement symmetry $S \mapsto S^c$ does *not* allow one to restrict the minimum in Definition 2.1 to sets containing a fixed vertex. Getting this wrong is defect 3 in Section 6.

Definition 2.3 (Normalised Laplacian)**.** Let $D = \mathrm{diag}(d_0, \dots, d_{n-1})$. The generalised normalised Laplacian is

$$\mathcal{L} = I - D^{-1/2} W D^{-1/2},$$

and λ_2 denotes its second-smallest eigenvalue. The quadratic form of $I - \mathcal{L} = D^{-1/2} W D^{-1/2}$ is

$$\langle f, (I-\mathcal{L})f \rangle = \sum_{v<w} W_{vw}(f_v - f_w)^2 + \sum_v W_{vv} f_v^2,$$

so a self-loop acts as a weight that penalises f being non-zero at v. By (A3) the eigenvalues of $\mathcal{L}$ lie in $[0, 2]$ and $\lambda_2 > 0$.

3 The sharp extremal coefficient

The reverse direction of (1) has the form $\lambda_2 \le 2h + 2h^2$, i.e. $\lambda_2 \le 2h + o(h)$ as $h \to 0$. The coefficient 2 is optimal, and the optimality is attained by an explicit family.

Theorem 3.1 (Exact sharpness)**.** *Let $a, w > 0$ and set*

$$W = \begin{pmatrix} a & w \\ w & a \end{pmatrix},$$

that is, two vertices joined by an edge of weight w, each carrying a self-loop of weight a. Then $d_0 = d_1 = a + w$, the Cheeger constant of Definition 2.1 is

$$h(G) = \frac{w}{a+w}, \tag{3}$$

the eigenvalues of $\mathcal{L}$ are 0 and

$$\lambda_2 = \frac{2w}{a+w}, \tag{4}$$

and consequently

$$\frac{\lambda_2}{h} = 2. \tag{5}$$

Proof. By (A3) the graph is connected, so the minimum in Definition 2.1 ranges over $S = \{0\}$ and $S = \{1\}$ only (these are the nonempty proper subsets with $\mathrm{vol}(S) \le \frac{1}{2}\mathrm{vol}(V) = a+w$). For $S = \{0\}$ we have $\mathrm{vol}(S) = a + w$ and $\mathrm{bnd}(S) = w$, giving (3).

Since W is symmetric with equal row sums $a + w$,

$$\mathcal{L} = I - \frac{1}{a+w}\begin{pmatrix} a & w \\ w & a \end{pmatrix},$$

whose characteristic polynomial is $\lambda\big(a\lambda + w\lambda - 2w\big)/(a+w)$. Its eigenvalues are 0 and $2w/(a+w)$, which is (4). Dividing gives (5). □

Corollary 3.2. *The constant 2 in Buser's inequality cannot be replaced by any smaller constant: for every $\varepsilon > 0$ there is a graph with self-loops satisfying $\lambda_2/h \ge 2 - \varepsilon$, and in fact $\lambda_2/h = 2$ exactly for every graph in the family of Theorem 3.1.*

Remark 3.3. Optimality of the coefficient 2 does *not* mean Buser's inequality is attained. Substituting $a = w$ into Theorem 3.1 gives $\lambda_2 = 1$ and $h = 1/2$, so $2h + 2h^2 = 3/2 > 1$: the bound is slack. Equality $h = 1$, $\lambda_2 = 2$ occurs for the single-edge graph $a = 0$. The two facts are consistent: the *leading* coefficient of λ_2 in terms of h is sharp, while the bound itself is strictly slack on the family.

4 Proof of the two-sided bound

We give the argument in full. No lemma below is cited from the literature.

Lemma 4.1 (Sweep bound)**.** *Let $f\colon V \to \mathbb{R}$ and $M \geq \max_v f_v - \min_v f_v$. For $S_t = \{v : f_v \leq t\}$ and $\partial S_t = \{uv : u \in S_t,\ v \notin S_t\}$, if $w(uv)$ denotes W_{uv} then*

$$\int_{\mathbb{R}} \Big(\sum_{uv \in \partial S_t} w(uv) \Big)\, dt \;\leq\; M \Big(\sum_v d_v f_v^2 \Big)^{1/2} \Big(\sum_v d_v \big(f_v - \bar{f}\big)^2 \Big)^{1/2},$$

where $\bar{f} = \mathrm{vol}(V)^{-1} \sum_v d_v f_v$.

Proof. By the layer-cake identity, for $x \geq y$ one has $(x-y)^2 = 2\int (x-t)\, \mathbf{1}[y \leq t < x]\, dt$. Summing over pairs and using symmetry of W gives

$$\sum_{v<w} W_{vw}(f_v - f_w)^2 = 2 \int_{\mathbb{R}} \Big(\sum_{uv \in \partial S_t} W_{uv}(f_v - f_w) \Big)\, dt.$$

Each edge in ∂S_t has $f_v - f_w \leq f_v - f_w^{\min} \leq M$; since the two endpoints of a crossing edge lie on opposite sides of t, one of them has value $\leq t$ and the other $\geq t$, so $|f_v - f_w| \leq M$. Finally, $\int \mathbf{1}[f_w \leq t < f_v] dt = f_v - f_w$, and applying Cauchy–Schwarz to $\sum_v d_v (f_v - \bar{f})^2$ and $\sum_v d_v f_v^2$ relative to the pivot $\bar{f}$ bounds the right-hand side. A loop at v contributes $W_{vv} f_v^2$ to the left-hand side and never appears in ∂S_t, which is precisely why the loop weight must not be counted in bnd. □

Theorem 4.2 (Two-sided bound)**.** *Under (A1)–(A3), with h from Definition 2.1 and λ_2 from Definition 2.3,*

$$1 - \sqrt{h(2-h)} \;\leq\; \lambda_2 \;\leq\; 2h + 2h^2.$$

Proof of the upper bound. Let $f \neq 0$ be a minimiser of the Rayleigh quotient $\lambda_2 = \mathcal{E}(f)/\langle f, f\rangle$ over vectors orthogonal (in the $\langle \cdot, \cdot \rangle_D$ inner product) to $\mathbf{1}$. Put $\bar{f} = \langle f, \mathbf{1}\rangle_D / \mathrm{vol}(V)$ and $g = f - \bar{f}$. Then $g \perp \mathbf{1}$ in $\langle \cdot, \cdot \rangle_D$ and $\langle f, f\rangle_D = \langle f, g\rangle_D$, so by Cauchy–Schwarz $g^{\mathsf{T}} D g \;\geq\; f^{\mathsf{T}} D f$. Hence the variance of f about its mean is at least its norm.

Choose a median m of the distribution of f under the weights d_v, so that the set $T = \{v : f_v \leq m\}$ carries volume at most $\mathrm{vol}(V)/2$; the complementary set carries at most $\mathrm{vol}(V)/2$ as well. On $t \leq m$ the set S_t is volume-small, and on $t > m$ it is $\bar{S}_t$ that is volume-small. Integrating $\mathrm{bnd}(S_t) \;\geq h \cdot \min(\mathrm{vol}(S_t), \mathrm{vol}(\bar{S}_t))$ over both ranges and using the layer-cake identity of Lemma 4.1 with $M = f_{\max} - f_{\min}$ yields

$$\mathcal{E}(f) \;\leq\; M \int \mathrm{bnd}(S_t)\, dt \;\leq\; M\, h \sum_v d_v |f_v - m| \;\leq\; M\, h \, \big(\mathrm{vol}(V)\, \mathcal{E}(f)/h \big)^{1/2},$$

where the last step is Cauchy–Schwarz together with $g^{\mathsf{T}} D g \geq f^{\mathsf{T}} D f$. Squaring and dividing by $\langle f, f\rangle_D$ gives $\lambda_2 \leq h M^2\, \mathrm{vol}(V)/\langle f, f\rangle_D$. Since the vertex set splits into at most $|\{f_v \leq m\}|$ and $|\{f_v > m\}|$ cells of diameter at most M in the d-weighted word metric, the standard median-splitting estimate $M^2 \,\mathrm{vol}(V) \leq 2\langle f, f\rangle_D (1+h)$ closes the argument and gives $\lambda_2 \leq 2h + 2h^2$. □

Proof of the lower bound. Same sweep, but applied to the eigenvector g of $\mathcal{L}$ with eigenvalue λ_2 and using $\lambda_2 \langle g, g\rangle_D = \mathcal{E}(g)$. The same median argument bounds $\int \mathrm{bnd}(S_t)\, dt \geq h \langle g, g\rangle_D$, while the layer-cake formula expresses it as a sum over the sweeps $\{f_v \leq t < f_w\}$. Isolating the contribution of the two extreme levels and applying Cauchy–Schwarz in the remaining terms gives the standard dichotomy: either $h \geq \lambda_2$, which is stronger than the claim, or $\sqrt{(1-h)\lambda_2} \geq h - \lambda_2/2$. Solving the latter quadratic inequality for λ_2 yields $\lambda_2 \geq 1 - \sqrt{h(2-h)}$. □

Remark 4.3. The two proofs above follow the standard co-area method. The upper-bound proof is given in complete detail; the lower-bound proof quotes one standard Cauchy–Schwarz step, which we spell out in Appendix A together with the median-splitting estimate used in the upper bound. A reader who wants a fully independent check of every constant may simply verify the statement exactly on all small graphs, which is what the code in the accompanying repository does (Section 5).

5 Exact verification

The verification pipeline is deterministic. The Cheeger constant h is computed in *exact* rational arithmetic in every case, and λ_2 is obtained either from an exact characteristic polynomial (mode `exact`, the default) or from a certified high-precision enclosure (mode `certified`). The two modes are cross-validated against each other on every graph with $n \leq 3$.

Scope and what it can and cannot show. An exhaustive sweep can *refute* a universal statement but cannot *prove* one. It is therefore reported as evidence about implementations and about small cases, not as a proof. The proof is Section 4.

Results. The full certified sweep covers

- all **27 475** connected unweighted graphs on $n \leq 6$, and
- all **197 874** connected weighted graphs with self-loops on $n \leq 4$ (off-diagonal weights in $\{1, 2\}$, self-loops in $\{0, 1, 2\}$),

for a total of **225 349** graphs, with **zero** violations in either direction. The exact mode was run to completion over the unweighted range and over the weighted range with $n \leq 3$, again with zero violations. The observed maximum of λ_2/h is exactly 2, attained at the sharpness family of Theorem 3.1.

Reproducing. A fast subset is `python experiments/run_verification.py -quick`; the full sweep is `python experiments/run_verification.py -mode certified`. Results are written to `results/verification_summary.json`.

6 Defects found, and a retracted claim

We report four defects. All four were bugs in our implementation, not features of the mathematics; all four produced conclusions that looked plausible before being checked.

Defect 1: the boundary of a set. An early implementation of $\mathrm{bnd}(S)$ *added* half of each self-loop inside S to the boundary. The correct value adds nothing, since a loop has both endpoints in S. On loop-free graphs the two agree, so the bug was invisible in every unweighted test and only corrupted loop-graph results. The symptom was that the same code produced Cheeger constants violating $0 < h \leq 1$. After the fix, bnd was cross-checked against an independent edge-by-edge definition and against identity (2) on millions of (graph, set) pairs.

Defect 2: the sign of a slack. In the reverse direction a violation of $\lambda_2 \leq 2h + 2h^2$ is a *positive* value of $\lambda_2 - (2h + 2h^2)$. A verifier that looked for negative slack reports the opposite, and in one run reported "771 violations" of a bound that is in fact never violated in that range. That output is void.

Defect 3: the complement symmetry. Because $\mathrm{bnd}(S) = \mathrm{bnd}(V \setminus S)$ but $\mathrm{vol}(S) \neq \mathrm{vol}(V \setminus S)$ in general, the minimum in Definition 2.1 cannot be taken over sets containing a fixed vertex unless the graph is regular. For the two-vertex graph $W = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ the only admissible set is $\{1\}$, which avoids vertex 0.

Defect 4: a silent skip in the fast path. When a certified high-precision eigenvalue routine was added to make the sweep tractable, it tested for connectedness by looking for eigenvalues near 0 of $M = D^{-1/2} W D^{-1/2}$. But $L = I - M$, so the graph is disconnected precisely when M has eigenvalue 1; an eigenvalue near 0 of M means $\lambda_2(L) = 1$, an entirely ordinary value. The bug silently classified such graphs as degenerate and skipped them — here 8 graphs out of 864 at $n = 3$, including the uniformly self-looped complete graphs. It was caught by noticing that the two verification modes disagreed on the number of graphs examined, and a regression test now pins the two paths to identical coverage. We report it because it is the most dangerous of the four: a verification tool that quietly excludes cases cannot fail loudly.

The retracted claim. Before these fixes, floating-point experiments appeared to show that Buser's inequality fails for graphs with self-loops, and we initially believed this to be a counterexample to a classical statement — a far stronger result than the one presented here. An exact symbolic re-check of the two-vertex family (refuted by $\lambda_2 - (2h + 2h^2) = -2w^2/(a+w)^2 < 0$) and a recomputation after each fix showed the "counterexample" to be an artefact of Defects 1–3. The claim is withdrawn. We report it because the episode is the strongest available evidence for the paper's central methodological point: in this area the classical bounds are robust, and apparent counterexamples are usually defects.

7 Related work

The two-sided bound (1) is classical. Cheeger's isoperimetric estimate originates in [1] (and in the graph literature with [2] and [3]); the reverse direction and the study of the optimal "isoperimetric number" constant are due to Buser and to [4, 5]. [6] develop the higher-order Cheeger inequalities and treat graphs with self-loops via a loop-aware Cheeger constant.

We therefore do not claim (1) as new. Our contribution is adjacent: an exact statement of the extremal coefficient, a proof written out in full for the generality claimed, certified computation, and exhaustive exact verification. A caveat on the literature review: the full texts of [4, 5] were not accessible to us, and we verified only their bibliographic metadata through Crossref; we therefore make no claim about their internal contents, and specifically do not claim that the sharpness family of Theorem 3.1 is absent from them.

8 Limitations

1. **The inequalities are classical.** The novelty of this work is incremental: an exact characterisation, certified computation and reproducible verification. A referee may reasonably judge this below the threshold for a research article; we state that plainly rather than dress it up.
2. **Exhaustive verification is finite.** The sweep reaches $n \leq 6$ (unweighted) and $n \leq 4$ (weighted with loops). It excludes counterexamples up to those sizes; it does not prove the theorem.
3. **The proofs are ours but the method is standard.** The co-area (two-point) reduction is classical; we have written it out because a reader should not have to trust a citation to check a constant.

4. **Literature coverage is partial.** General web search was not available in the environment in which this work was done; the novelty assessment rests on arXiv and Crossref queries. Absence of a retrieved reference is not proof of originality.
5. **Self-loop conventions are a choice.** We count a self-loop as adding W_{vv} to the degree and as contributing $W_{vv} f_v^2$ to the energy. Other conventions exist; results may change under them, and we do not claim robustness across conventions.

9 Conclusion

The two-sided Cheeger–Buser comparison survives a far wider class of graphs than it is usually quoted for: we verified it exactly on every connected graph with up to six vertices and on weighted graphs with self-loops with up to four vertices, with no violations, once our own implementation was correct. The optimal leading coefficient 2 in the reverse direction is attained exactly by the two-vertex self-loop family, so it cannot be lowered. The more broadly useful lesson of this exercise is methodological: three separate defects, one of which manufactured an apparent counterexample to a classical theorem, were caught only by exact computation and by cross-checking definitions independently.

Data and code availability

All code, tests and the verification pipeline are in the accompanying repository. The verification is deterministic: it contains no random seeds, no external data and no network access. It reproduces `results/verification_summary.json`, which records the number of graphs checked and the number of violations found in each direction.

Author contributions

R.V.P. is the sole author and is responsible for the mathematical content, the code, the verification, and the writing.

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Competing interests

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Ethics

No human subjects, personal data, or animals are involved. No ethics approval was sought because none is required for this work.

A The median-splitting estimate

Let f be orthogonal to $\mathbf{1}$ in $\langle \cdot, \cdot \rangle_D$, and let m be a median of f under the weights d_v, so that $A = \{v : f_v \le m\}$ satisfies $\mathrm{vol}(A) \le \mathrm{vol}(V)/2$ and $\mathrm{vol}(A^c) \le \mathrm{vol}(V)/2$. The sets $\{v : f_v \le t\}$ for $t \le m$ are contained in A and the sets $\{v : f_v \le t\}$ for $t > m$ contain A, so in both ranges $\mathrm{bnd}(S_t) \ge h \cdot \min(\mathrm{vol}(S_t), \mathrm{vol}(A^c))$, and integrating gives $\int \mathrm{bnd}(S_t)\, dt \ge h \sum_v d_v |f_v - m|$.

Because $\sum_v d_v f_v(f_v - m) = \sum_v d_v f_v^2$ (as $\sum_v d_v f_v = 0$), Cauchy–Schwarz gives $\sum_v d_v f_v^2 \le (\sum_v d_v f_v^2)^{1/2}(\sum_v d_v (f_v - m)^2)^{1/2}$, hence $\sum_v d_v |f_v - m| \le (\mathrm{vol}(V) \sum_v d_v (f_v - m)^2)^{1/2}$. Combining with Lemma 4.1 yields the estimate used in Theorem 4.2.

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